1 Reachability and controllability

1.1 Reachable systems

Definition 1. *Given a continuous-time system*

$$\dot{x} \;=\; A\,x + B\,u \qquad x \in \mathbb{R}^n, u \in \mathbb{R}^m$$

*the state $\bar{x}$ is **reachable** if $\exists \bar{t}, \bar{u}$ such that*

$$\bar{x} \;=\; \int_0^{\bar{t}} e^{A(\bar{t}-\tau)} B \bar{u}(\tau)\, d\tau$$

Theorem 2. *For the continuous-time system*

$$\dot{x} \;=\; A\,x + B\,u \qquad x \in \mathbb{R}^n, u \in \mathbb{R}^m$$

$\bar{x}$ is reachable $\Leftrightarrow$

$$\bar{x} \;\in\; \mathrm{Im}(P)$$

*where P is the **reachability matrix** given by*

$$P \;=\; \begin{pmatrix} B & A\,B & \dots & A^{n-1}B \end{pmatrix}$$

The system is reachable $\Leftrightarrow$

$$\mathrm{rk}\, P \;=\; n$$

Theorem 3. *If the LTI system*

$$\dot{x} \;=\; A\,x + B\,u \qquad x \in \mathbb{R}^n, u \in \mathbb{R}^m$$

is reachable, then $\exists K$ such that the closed-loop system

$$\dot{x} \;=\; (A + B\,K)\,x$$

is asymptotically stable.

1.2 Controllable systems

Definition 4. *Given a continuous-time system*

$$\dot{x} \;=\; A\,x + B\,u \qquad x \in \mathbb{R}^n, u \in \mathbb{R}^m$$

*the state $\bar{x}$ is **controllable** if $\exists \bar{t}, \bar{u}$ such that*

$$0 \;=\; e^{A\bar{t}} x_0 + \int_0^{\bar{t}} e^{A(\bar{t}-\tau)} B \bar{u}(\tau)\, d\tau$$

Theorem 5. *In a continuous-time system, a state $\bar{x}$ is reachable if and only if it is controllable.*

Corollary 6. *A system is reachable $\Leftrightarrow$ it is controllable.*

2 Full-information regulator problem

Consider the LTI system

$$\begin{aligned} \dot{x} &= A\,x + B\,u + P\,d \\ e &= C\,x + Q\,d \end{aligned}$$

where the exogenous signal $d(t)$, which represents both disturbances and references, has dynamics given by

$$\dot{d} = S\,d$$

Assume that the state $x(t)$ and the exogenous signal $d(t)$ are measurable, and consider the state-feedback control law

$$u = K\,x + L\,d$$

which gives

$$\begin{aligned} \dot{x} &= A\,x + B\,u + P\,d \\ &= A\,x + B\,(K\,x + L\,d) + P\,d \\ &= A\,x + B\,K\,x + B\,L\,d + P\,d \\ &= (A + B\,K)x + (B\,L + P)d \\ e &= C\,x + Q\,d \end{aligned}$$

The **full-information regulator problem** is to determine the matrices K and L such that

- the closed-loop system in the absence of the exogenous signal ($d(t) = 0$)

  $$\dot{x} = (A + B\,K)x$$

  is asymptotically stable
- the tracking error converges asymptotically to zero for every initial condition.

  $$\lim_{t\to\infty} \dot{e}(t) = 0$$

Theorem 7.

Consider a full-information regulator problem in which

- *the exogenous signal $d(t)$ persists over time, so that*

  $$\mathrm{Re}(\lambda_i) \geq 0 \qquad \forall \lambda_i \in \sigma(S)$$

- *the system is stabilisable.*

Then the tracking error $e(t)$ converges asymptotically to zero ($\lim_{t\to\infty} e(t) = 0$) $\Leftrightarrow$ $\exists$ a matrix Π such that

$$\begin{aligned} \Pi\,S &= (A + B\,K)\Pi + (B\,L + P) \\ 0 &= C\,\Pi + Q \end{aligned}$$

Proof.

Since the equation

$$\begin{aligned} \Pi S &= (A+BK)\Pi + (BL+P) \\ \Pi S - (A+BK)\Pi &= BL+P \end{aligned}$$

is a Sylvester equation, it has a unique solution if and only if

$$\sigma(S) \cap \sigma(A+BK) = \emptyset$$

Since $\sigma(s) \in \mathbb{C}^+$ and $\sigma(A+BK) \in \mathbb{R}^-$ by assumption, the matrix Π exists and is unique.

Define

$$\hat{x} = x - \Pi d$$

to obtain

$$\begin{aligned} \dot{\hat{x}} &= \dot{x} - \Pi \dot{d} \\ &= (A+BK)x + (BL+P)d - \Pi S d \\ &= (A+BK)x + (BL+P-\Pi S)d \end{aligned}$$

Since Π satisfies the Sylvester equation,

$$BL+P-\Pi S = -(A+BK)\Pi$$

it follows that

$$\begin{aligned} \dot{\hat{x}} &= (A+BK)x - (A+BK)\Pi d \\ &= (A+BK)(x-\Pi d) \\ &= (A+BK)\hat{x} \end{aligned}$$

Therefore, $\hat{x}$ converges asymptotically to zero, which gives

$$\begin{aligned} \lim_{t\to\infty} \hat{x} &= 0 \\ x &\to \Pi d \end{aligned}$$

Substituting into the tracking error expression gives

$$\begin{aligned} e &= Cx + Qd \\ &\to C\Pi d + Qd \\ &= (C\Pi + Q)d \end{aligned}$$

The tracking error e therefore converges asymptotically to zero if and only if

$$C\Pi + Q = 0$$

This proves the result. □

Theorem 8.

Consider a full-information regulator problem in which

- *the exogenous signal $d(t)$ persists over time, so that*

$$\mathrm{Re}(\lambda_i) \;\geq\; 0 \qquad \forall \lambda_i \in \sigma(S)$$

- *the system is stabilisable.*

Then the tracking error $e(t)$ converges asymptotically to zero ($\lim_{t\to\infty} e(t) = 0$) $\Leftrightarrow$ $\exists$ matrices Π and Γ such that

$$\begin{aligned} \Pi S &= A\Pi + B\Gamma + P \\ 0 &= C\Pi + Q \end{aligned}$$

Proof.

By Theorem 7, the full-information regulator problem admits a solution $\Leftrightarrow$ $\exists\, \Pi$ such that

$$\begin{aligned} \Pi S &= (A + BK)\Pi + BL + P \\ 0 &= C\Pi + Q \end{aligned}$$

where

$$\begin{aligned} \Pi S &= (A + BK)\Pi + BL + P \\ &= A\Pi + BK\Pi + BL + P \\ &= A\Pi + B(K\Pi + L) + P \end{aligned}$$

Setting $\Gamma = K\Pi + L$ proves the result. □

Theorem 9. ***(Hautus's lemma)***

If $x \in \mathbb{R}^n, e \in \mathbb{R}^p$, the full-information regulator problem is solvable for any choice of P and Q if and only if $\forall \lambda \in \sigma(S)$

$$\mathrm{rank}\begin{bmatrix} \lambda I - A & B \\ C & 0 \end{bmatrix} = n + p$$

3 Error-feedback regulator problem

Consider the LTI system

$$\begin{aligned} \dot{x} &= A\,x + B\,u + P\,d \\ e &= C\,x + Q\,d \end{aligned}$$

where the exogenous signal $d(t)$, which represents disturbances and references, has dynamics given by

$$\dot{d} = S\,d$$

Assume that only the tracking error $e(t)$ can be measured, and consider the dynamic controller

$$\begin{aligned} \dot{\chi} &= F\chi + G\,e \\ u &= H\chi \end{aligned}$$

The **error-feedback regulator problem** is to determine the matrices F, G, and H such that

- the closed-loop system in the absence of the exogenous signal ($d = 0$)

$$\begin{aligned} \dot{x} &= A\,x + B\,u \\ \dot{\chi} &= F\chi + G\,e \end{aligned}$$

  is asymptotically stable

- the tracking error converges asymptotically to zero for every initial condition.

$$\lim_{t\to\infty} e(t) = 0.$$

Theorem 10.

Consider an error-feedback regulator problem in which

- *the exogenous signal $d(t)$ persists over time, so that*

$$\sigma(S) \in \mathbb{C}^-$$

- *the system*

$$\dot{x} = A\,x + B\,u + P\,d$$

  is stabilisable

- *the system*

$$\begin{aligned} \begin{pmatrix} \dot{x} \\ \dot{d} \end{pmatrix} &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix} u \\ e &= \begin{pmatrix} C & Q \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix} \end{aligned}$$

is observable.

Then the problem is solvable if and only if $\exists\ \Pi,$ *and* Γ *such that*

$$\begin{aligned} \Pi S &= A\Pi + B\Gamma + P \\ 0 &= C\Pi + Q \end{aligned}$$

Remark 11. If the three assumptions of the theorem hold, the error-feedback regulator problem is solvable if and only if the full-information regulator problem is solvable.

Proof.

$\Rightarrow$

The proof of necessity is the same as for the full-information regulator problem.

$\Leftarrow$

Let $\hat{x}(t)$ be the estimate of $x(t)$ and $\hat{d}(t)$ be the estimate of $d(t)$, with

$$\begin{aligned} \begin{pmatrix} \dot{\hat{x}} \\ \dot{\hat{d}} \end{pmatrix} &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix}u + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}(\hat{e} - e) \\ \hat{e} &= \begin{pmatrix} C & Q \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix} \end{aligned}$$

If

$$e_x = \hat{x} - x$$

is the estimation error for x and

$$e_d = \hat{d} - d$$

is the estimation error for d, then

$$\begin{aligned} \begin{pmatrix} \dot{e}_x \\ \dot{e}_d \end{pmatrix} &= \begin{pmatrix} \dot{\hat{x}} - \dot{x} \\ \dot{\hat{d}} - \dot{d} \end{pmatrix} \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix}u + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}(\hat{e} - e) - \left(\begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix}u\right) \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix} + \begin{pmatrix} \cancel{B} \\ \cancel{\emptyset} \end{pmatrix}u + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}(\hat{e} - e) - \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix} - \begin{pmatrix} \cancel{B} \\ \cancel{0} \end{pmatrix}u \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix} - \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}(\hat{e} - e) \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} \hat{x} - x \\ \hat{d} - d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}(\hat{e} - e) \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} e_x \\ e_d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}(\hat{e} - e) \end{aligned}$$

Recall that

$$\begin{aligned}\hat{e} &= \begin{pmatrix} C & Q \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{d} \end{pmatrix} = C\hat{x} + Q\hat{d} \\ e &= \begin{pmatrix} C & Q \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix} = Cx + Qd\end{aligned}$$

so

$$\begin{aligned}\begin{pmatrix} \dot{e}_x \\ \dot{e}_d \end{pmatrix} &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} e_x \\ e_d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}(\hat{e} - e) \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} e_x \\ e_d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}[C\hat{x} + Q\hat{d} - (Cx + Qd)] \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} e_x \\ e_d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}[C(\hat{x} - x) + Q(\hat{d} - d)] \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} e_x \\ e_d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}[Ce_x + Qe_d] \\ &= \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} e_x \\ e_d \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}\begin{pmatrix} C & Q \end{pmatrix}\begin{pmatrix} e_x \\ e_d \end{pmatrix} \\ &= \left[\begin{pmatrix} A & P \\ 0 & S \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}\begin{pmatrix} C & Q \end{pmatrix}\right]\begin{pmatrix} e_x \\ e_d \end{pmatrix}\end{aligned}$$

Since the pair $\left(\begin{smallmatrix} A & P \\ 0 & S \end{smallmatrix}\right), \begin{pmatrix} C & Q \end{pmatrix}$ is observable, the estimation error dynamics can be stabilised. In particular, there exist gains G_1, G_2 such that the matrix

$$\begin{pmatrix} A & P \\ 0 & S \end{pmatrix} + \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}\begin{pmatrix} C & Q \end{pmatrix}$$

is Hurwitz.

In the full-information regulator problem, $u = Kx + Ld$. Replace x with its estimate $\hat{x}$ and d with its estimate $\hat{d}$. Recalling that

$$\begin{aligned} e_x = \hat{x} - x \quad &\Rightarrow \quad \hat{x} = e_x - x \\ e_d = \hat{d} - d \quad &\Rightarrow \quad \hat{d} = e_d - d \end{aligned}$$

we obtain

$$\begin{aligned} u &= -K\hat{x} - L\hat{d} \\ &= -K(e_x - x) - L(e_d - d) \\ &= -Ke_x + Kx - Le_d + Ld \\ &= Kx + Ld - Ke_x - Le_d \end{aligned}$$

Since $e_x \to 0$ and $e_d \to 0$, we have

$$u \;\to\; Kx + Ld$$

Asymptotically, the problem therefore reduces to a full-information regulator problem.

□

4 Assignment 7

Consider the model of an **inverted pendulum on a cart**, described by the equations

$$\begin{aligned} M\ddot{s} + F\dot{s} - \mu &= d_1 \\ \ddot{\phi} - \frac{g}{L}\sin\phi + \frac{1}{L}\ddot{s}\cos\phi &= 0 \end{aligned}$$

where

- $s(t)$ is the **displacement of the suspension point** (the cart)
- $\phi(t)$ is the **angular displacement of the pendulum**
- $\mu(t)$ is the **external force applied to the cart** (the control input)
- $d_1(t)$ is an **external disturbance** acting on the cart.

The parameters are

- $M = 1$ kg, the mass of the cart
- $L = 1\ m$, the effective length of the pendulum
- $F = 1\ \frac{\text{kg}}{s}$, the viscous friction coefficient
- $g = 9.81\ \frac{m}{s^2}$, the acceleration due to gravity.

1. Find all equilibrium points of the system when

$$\mu(t) = 0, \qquad d_1(t) = 0$$

2. Write the equations of the system linearised about the equilibrium point

$$\phi = s = \dot{\phi} = \dot{s} = 0$$

3. Express the resulting linear dynamics in standard state-space form,

$$\begin{aligned} \dot{x} &= A\,x + B\,u + P\,d_1 \\ y &= C\,x \end{aligned}$$

where

$$x(t) = \begin{bmatrix} s(t) \\ \dot{s}(t) \\ \phi(t) \\ \dot{\phi}(t) \end{bmatrix}$$

and

$$y(t) = \begin{bmatrix} s(t) \\ \phi(t) \end{bmatrix}$$

4. Show that the pair (A, B) is controllable.

5. Consider the linear system obtained in Step 3.

   Assume that d_1 is an unknown constant.

   Design the control law to meet the following requirements.

   - The effect of the disturbance d_1 is asymptotically rejected.
   - The first output $s(t)$ asymptotically tracks the reference signal

   $$d_2(t) = \alpha \sin(\omega t).$$

   Formulate this as a regulation problem.

   Note that the regulation condition applies only to part of the measured output, which makes this a non-standard regulator problem.

6. Consider the regulation problem defined in Step 5.

   Show that it can be solved using a full-information control law.

7. Return to the regulation problem in Step 5.

   Show that it can be solved using an error-feedback control law.

   Because the problem differs from the standard form discussed in class, use the output $y(t)$ to verify observability and use the error signal

   $$e(t) \;=\; s(t) - d_2(t)$$

   to check whether the regulation requirement is satisfied.

Also complete the following steps.

i. Assume that

   - $d_1(t)$ is a square wave of amplitude 0.5 and period $50s$
   - $\alpha = 1$
   - $\omega = 0.1$

   Design a full-information control law that solves the regulation problem formulated in Step 5.

ii. Plot $y(t)$ and $u(t)$ for the initial condition $x(0) = 0$.

   The plots should show that the regulation objective has been achieved.

   Explain why the disturbance $d_1(t)$ does not affect the output $y(t)$.

iii. Now use the control law designed in Step i to control the nonlinear pendulum model.

   Plot $y(t)$ and $u(t)$ for the nonlinear system with $x(0) = 0$.

   Is the regulation objective still satisfied?

   Explain the following observations.

   - The disturbance $d_1(t)$ does not affect the output $y(t)$.
   - The regulation objective is achieved only approximately.

iv. Repeat Steps i to iii for

   $$\omega = 1$$

   and

   $$\omega = 10.$$

   Discuss the results.

5 Linearisation

5.1 Equilibrium points

To find the equilibrium points of

$$\begin{aligned} M\ddot{s} + F\dot{s} - \mu &= d_1 \\ \ddot{\phi} - \frac{g}{L}\sin\phi + \frac{1}{L}\ddot{s}\cos\phi &= 0 \end{aligned}$$

with $\mu, d_1 = 0$, set

$$\begin{aligned} \dot{s} &= 0 \\ \ddot{s} &= 0 \\ \dot{\phi} &= 0 \\ \ddot{\phi} &= 0 \end{aligned}$$

to obtain

$$\begin{aligned} 0 &= 0 \\ -\frac{g}{L}\sin\phi &= 0 \end{aligned}$$

The equilibrium points of the system are therefore

$$\begin{pmatrix} s_e & \dot{s}_e & \phi_e & \dot{\phi}_e \end{pmatrix} = \begin{pmatrix} s & 0 & k\pi & 0 \end{pmatrix}$$

5.2 Linearisation

To linearise the system about the point

$$\begin{pmatrix} s_e & \dot{s}_e & \phi_e & \dot{\phi}_e \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$$

note the following.

- The first equation

  $$M\ddot{s} + F\dot{s} - \mu = d_1$$

  is already linear.

- Since

  $$\begin{aligned} \sin\phi &\approx \phi \\ \cos\phi &\approx 1 \end{aligned}$$

  the second equation becomes

  $$\ddot{\phi} - \frac{g}{L}\phi + \frac{1}{L}\ddot{s} = 0$$

The linearised system is therefore

$$\begin{aligned} M\ddot{s} + F\dot{s} - \mu &= d_1 \\ \ddot{\phi} - \frac{g}{L}\phi + \frac{1}{L}\ddot{s} &= 0 \end{aligned}$$

5.3 Linear system

Write the linearised system in standard form by defining

$$x(t) = \begin{bmatrix} s(t) \\ \dot{s}(t) \\ \phi(t) \\ \dot{\phi}(t) \end{bmatrix}$$

and

$$y(t) = \begin{bmatrix} s(t) \\ \phi(t) \end{bmatrix}$$

Since

$$\begin{aligned} M\ddot{s} + F\dot{s} - \mu &= d_1 \\ \ddot{\phi} - \frac{g}{L}\phi + \frac{1}{L}\ddot{s} &= 0 \end{aligned}$$

$$\begin{aligned} M\ddot{s} &= -F\dot{s} + \mu + d_1 \\ \ddot{\phi} &= \frac{g}{L}\phi - \frac{1}{L}\ddot{s} \end{aligned}$$

$$\begin{aligned} \ddot{s} &= \frac{-F\dot{s} + \mu + d_1}{M} \\ \ddot{\phi} &= \frac{g}{L}\phi - \frac{1}{L}\ddot{s} \end{aligned}$$

$$\begin{aligned} \ddot{s} &= -\frac{F}{M}\dot{s} + \frac{1}{M}\mu + \frac{1}{M}d_1 \\ \ddot{\phi} &= \frac{g}{L}\phi - \frac{1}{L}\ddot{s} \end{aligned}$$

we obtain

$$\begin{aligned} \dot{x}_1 &= \dot{s} \\ &= x_2 \\ \dot{x}_2 &= \ddot{s} \\ &= -\frac{F}{M}\dot{s} + \frac{1}{M}\mu + \frac{1}{M}d_1 \\ &= -\frac{F}{M}x_2 + \frac{1}{M}\mu + \frac{1}{M}d_1 \\ \dot{x}_3 &= \dot{\phi} \\ &= x_4 \\ \dot{x}_4 &= \ddot{\phi} \\ &= \frac{g}{L}\phi - \frac{1}{L}\ddot{s} \end{aligned}$$

$$
\begin{aligned}
&= \frac{g}{L}x_3 - \frac{1}{L}\dot{x}_2 \\
&= \frac{g}{L}x_3 - \frac{1}{L}\left(-\frac{F}{M}x_2 + \frac{1}{M}\mu + \frac{1}{M}d_1\right) \\
&= \frac{g}{L}x_3 + \frac{F}{L\,M}x_2 - \frac{1}{L\,M}\mu - \frac{1}{L\,M}d_1
\end{aligned}
$$

In matrix form, this gives

$$
\begin{aligned}
\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} &= \begin{pmatrix} x_2 \\ -\frac{F}{M}x_2 + \frac{1}{M}\mu + \frac{1}{M}d_1 \\ x_4 \\ \frac{g}{L}x_3 + \frac{F}{L\,M}x_2 - \frac{1}{L\,M}\mu - \frac{1}{L\,M}d_1 \end{pmatrix} \\
&= \begin{pmatrix} x_2 \\ -\frac{F}{M}x_2 \\ x_4 \\ \frac{F}{L\,M}x_2 + \frac{g}{L}x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{M}\mu \\ 0 \\ -\frac{1}{L\,M}\mu \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{M}d_1 \\ 0 \\ -\frac{1}{L\,M}d_1 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{F}{M} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{F}{L\,M} & \frac{g}{L} & 0 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{L\,M} \end{pmatrix} \mu + \begin{pmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{L\,M} \end{pmatrix} d_1
\end{aligned}
$$

Recall that

- $M = 1$ kg
- $L = 1$ m
- $F = 1$ $\frac{\text{kg}}{s}$

so

$$
\dot{x} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & g & 0 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} u + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} d_1
$$

The output $y(t)$ is given by

$$
\begin{aligned}
y(t) &= \begin{pmatrix} s \\ \phi \end{pmatrix} \\
&= \begin{pmatrix} x_1 \\ x_3 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}
\end{aligned}
$$

In summary, the linearised system is

$$\begin{aligned}
\dot{x} &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & g & 0 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} u + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} d_1 \\
y &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}
\end{aligned}$$

5.4 Controllability

To verify that the pair (A, B) is controllable, check that the matrix

$$P = \begin{pmatrix} B & AB & A^2B & A^3B \end{pmatrix}$$

has full rank, where

$$\begin{aligned}
AB &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & g & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \\
&= \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} \\
A^2B &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & g & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1-g \end{pmatrix} \\
A^3B &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & g & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1-g \end{pmatrix} \\
&= \begin{pmatrix} 1 \\ -1 \\ -1-g \\ 1+g \end{pmatrix}
\end{aligned}$$

The reachability matrix is therefore

$$P = \begin{pmatrix} 0 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & -1 & 1 & -1-g \\ -1 & 1 & -1-g & 1+g \end{pmatrix}$$

Reducing it to column echelon form gives

$$\begin{pmatrix} 0 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & -1 & 1 & -1-g \\ -1 & 1 & -1-g & 1+g \end{pmatrix}$$

$$e^{24}(-1)\downarrow e^{34}(1)$$

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & g & -g & -1-g \\ -1 & -g & 0 & 1+g \end{pmatrix}$$

$$\downarrow p^{13}$$

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ -g & g & 0 & -1-g \\ 0 & -g & -1 & 1+g \end{pmatrix}$$

$$\downarrow e^{12}(1)$$

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & g & 0 & -1-g \\ -g & -g & -1 & 1+g \end{pmatrix}$$

Since this matrix has full rank, the pair (A, B) is controllable.

6 Regulation

6.1 Problem formulation

First, define the error dynamics.

Since

$$\begin{aligned} d_2' &= \frac{d}{\mathrm{dt}}(\alpha \sin(\omega t)) \\ &= \alpha\omega\cos(\omega t) \\ d_2'' &= \frac{d}{\mathrm{dt}}(\alpha\omega\cos(\omega t)) \\ &= -\alpha\,\omega^2 \sin(\omega\, t) \\ &= -\omega^2 d_2 \end{aligned}$$

define

$$d_3 = \alpha\omega\cos(\omega t)$$

to obtain

$$\begin{aligned} \dot{d}_2 &= d_3 \\ \dot{d}_3 &= -\omega^2 d_2 \end{aligned}$$

The error dynamics are therefore

$$\begin{aligned} \begin{pmatrix} \dot{d}_1 \\ \dot{d}_2 \\ \dot{d}_3 \end{pmatrix} &= \begin{pmatrix} 0 \\ d_3 \\ -\omega^2 d_2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -\omega^2 & 0 \end{pmatrix} \cdot \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} \end{aligned}$$

The dynamics of the state x can then be written as

$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & g & 0 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} u + \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} d$$

The tracking error is

$$\begin{aligned} e &= x_1 - d_2 \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} \end{aligned}$$

In summary, the regulation problem concerns the system

$$\begin{aligned} \dot{x} &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & g & 0 \end{pmatrix} \cdot x + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} u + \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} d \\ e &= \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} x + \begin{pmatrix} 0 & -1 & 0 \end{pmatrix} d \end{aligned}$$

$$\dot{d} \;=\; \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -\omega^2 & 0 \end{pmatrix} d$$

6.2 Full-information regulator problem

By Hautus's lemma, the regulation problem is solvable $\forall P, Q$ if and only if

$$\operatorname{rank}\begin{bmatrix} \lambda I - A & B \\ C & 0 \end{bmatrix} = 4+1=5 \quad \forall \lambda \in \sigma(S)$$

We therefore calculate the eigenvalues of the matrix S.

$$\begin{aligned} \det(\lambda I - S) \;&=\; \det\begin{pmatrix} s & 0 & 0 \\ 0 & s & -1 \\ 0 & \omega^2 & s \end{pmatrix} \\ &=\; s(s^2+\omega^2) \end{aligned}$$

It follows that

$$\sigma(S) \;=\; \{0, \pm j\omega\}$$

and

- For $\lambda = 0$,

$$\begin{pmatrix} \lambda I - A & B \\ C & 0 \end{pmatrix} \;=\; \begin{pmatrix} 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & -g & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

  reducing to row echelon form gives

$$\begin{pmatrix} 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & -g & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\downarrow p_{14}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & -g & 0 & -1 \\ 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

$$e_{42}(1)\downarrow e_{52}(1)$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -g & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Therefore, $\mathrm{rk}\begin{pmatrix} -A & B \\ C & 0 \end{pmatrix} = 5$.

- For $\lambda = j\omega$,

$$\begin{pmatrix} \lambda I - A & B \\ C & 0 \end{pmatrix} = \begin{pmatrix} j\omega & -1 & 0 & 0 & 0 \\ 0 & j\omega+1 & 0 & 0 & 1 \\ 0 & 0 & j\omega & -1 & 0 \\ 0 & -1 & -g & j\omega & -1 \\ 1 & 0 & 0 & 0 & j\omega \end{pmatrix}$$

reducing to row echelon form gives

$$\begin{pmatrix} j\omega & -1 & 0 & 0 & 0 \\ 0 & j\omega+1 & 0 & 0 & 1 \\ 0 & 0 & j\omega & -1 & 0 \\ 0 & -1 & -g & j\omega & -1 \\ 1 & 0 & 0 & 0 & j\omega \end{pmatrix}$$

$$\downarrow e_{51}\left(-\frac{1}{j\omega}\right)$$

$$\begin{pmatrix} j\omega & -1 & 0 & 0 & 0 \\ 0 & j\omega+1 & 0 & 0 & 1 \\ 0 & 0 & j\omega & -1 & 0 \\ 0 & -1 & -g & j\omega & -1 \\ 0 & \frac{1}{j\omega} & 0 & 0 & j\omega \end{pmatrix}$$

$$\downarrow p_{25}$$

$$\begin{pmatrix} j\omega & -1 & 0 & 0 & 0 \\ 0 & \frac{1}{j\omega} & 0 & 0 & j\omega \\ 0 & 0 & j\omega & -1 & 0 \\ 0 & -1 & -g & j\omega & -1 \\ 0 & j\omega+1 & 0 & 0 & 1 \end{pmatrix}$$

$$e_{42}(j\omega)\downarrow e_{52}(-j\omega(j\omega+1))$$

$$\begin{pmatrix} j\omega & -1 & 0 & 0 & 0 \\ 0 & \frac{1}{j\omega} & 0 & 0 & j\omega \\ 0 & 0 & j\omega & -1 & 0 \\ 0 & 0 & -g & j\omega & -1-\omega^2 \\ 0 & 0 & 0 & 0 & 1+\omega^2(j\omega+1) \end{pmatrix}$$

$$\downarrow e_{43}\left(\frac{g}{j\omega}\right)$$

$$\begin{pmatrix} j\omega & -1 & 0 & 0 & 0 \\ 0 & \frac{1}{j\omega} & 0 & 0 & j\omega \\ 0 & 0 & j\omega & -1 & 0 \\ 0 & 0 & 0 & j\omega - \frac{g}{j\omega} & -1-\omega^2 \\ 0 & 0 & 0 & 0 & \omega^2(j\omega+1) \end{pmatrix}$$

Therefore, $\mathrm{rk}\begin{pmatrix} j\omega - A & B \\ C & 0 \end{pmatrix} = 5 = \overline{\mathrm{rk}\begin{pmatrix} j\omega - A & B \\ C & 0 \end{pmatrix}}$.

The problem is therefore solvable for any choice of P and Q.

6.3 Error-feedback regulator problem

By Theorem 10, a solution to the error-feedback regulator problem exists if

$$\begin{pmatrix} \dot{x} \\ \dot{d} \end{pmatrix} = \begin{pmatrix} A & P \\ 0 & S \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix} u$$

$$e = \begin{pmatrix} C & Q \end{pmatrix}\begin{pmatrix} x \\ d \end{pmatrix}$$

is observable. In that case, the problem reduces to a full-information regulator problem.

We have

$$\begin{pmatrix} A & P \\ 0 & S \end{pmatrix} = \left(\begin{array}{cccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & g & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega^2 & 0 \end{array}\right)$$

$$\begin{pmatrix} C & Q \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}$$

Define

$$A_e = \begin{pmatrix} A & P \\ 0 & S \end{pmatrix} \qquad C_e = \begin{pmatrix} C & Q \end{pmatrix}$$

The observability matrix Q is then

$$Q = \left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 & 1 & \omega^2 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & \omega^2 \\ 0 & -1 & 0 & 0 & 1 & -\omega^4 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & -\omega^4 \\ 0 & -1 & 0 & 0 & 1 & \omega^6 & 0 \end{array}\right)$$

Since the third and fourth columns are zero, this matrix does not have full rank.

Since the angular displacement of the pendulum $x_3 = \phi$ is known, we can define the measurement error

$$e_M = \begin{pmatrix} x_1 - d_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} x + \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} d$$

The first seven rows of the observability matrix then become

$$Q_M = \left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & \omega^2 & 0 \\ 0 & 1 & g & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & \omega^2 \end{array}\right)$$

To determine the rank of Q_m, reduce it to row echelon form.

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & \omega^2 & 0 \\ 0 & 1 & g & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & \omega^2 \end{array}\right)$$

$$p_{23}\downarrow$$

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & \omega^2 & 0 \\ 0 & 1 & g & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & \omega^2 \end{array}\right)$$

$$e_{52}(1)\downarrow e_{62}(-1)e_{72}(-1)$$

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \omega^2 & -1 \\ 0 & 0 & g & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & \omega^2+1 \end{array}\right)$$

$$\downarrow e_{63}(-g)$$

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \omega^2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & \omega^2+1 \end{array}\right)$$

$$\downarrow e_{64}(1)$$

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \omega^2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & \omega^2 & \omega^2 \end{array}\right)$$

$$\downarrow p_{56}\left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \omega^2 & -1 \\ 0 & 0 & 0 & 0 & 0 & \omega^2 & \omega^2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

The observability matrix has full rank, so the error-feedback regulator problem is solvable.

7 Full-information regulator design

To solve the full-information regulator problem, first find a matrix K that stabilises the system.

$$\dot{x} \;=\; (A + B\,K)x$$

We have

$$\begin{aligned}
\det(s\,I - A) \;&=\; \det\begin{pmatrix} s & -1 & 0 & 0 \\ 0 & s+1 & 0 & 0 \\ 0 & 0 & s & -1 \\ 0 & -1 & -g & s \end{pmatrix} \\
&=\; s(s+1)(s^2 - g) \\
&\Downarrow \\
\sigma(A) \;&=\; \{0, -1, \pm\sqrt{g}\}
\end{aligned}$$

We therefore need to move the eigenvalues $\lambda_a = 0$ and $\lambda_b = \sqrt{g}$. In MATLAB, use *place* to impose

$$\sigma(A + B\,K) \;=\; \{-1, -\sqrt{g}, -1.2, -0.8\}$$

Next, find two matrices Π, Γ that satisfy the equations

$$\begin{aligned}
\Pi\,S \;&=\; A\,\Pi + B\,\Gamma + P \\
0 \;&=\; C_s\,\Pi + Q
\end{aligned}$$

where C_s is the matrix that relates the state to the tracking error e.

To do this, use the function

$$\text{generalized_sylvester}(A, B, C, D, E)$$

which solves the Sylvester equation

$$A\,X\,B + C\,X\,D \;=\; E$$

Rewrite the equations to be solved in terms of

$$X \;=\; \begin{pmatrix} \Pi \\ \Gamma \end{pmatrix}$$

We obtain

$$\begin{aligned}
\Pi\,S \;&=\; A\,\Pi + B\,\Gamma + P \\
0 \;&=\; C_s\,\Pi + Q
\end{aligned}$$

$$\begin{pmatrix} \Pi\,S \\ 0 \end{pmatrix} \;=\; \begin{pmatrix} A\Pi + B\,\Gamma \\ C_s\,\Pi \end{pmatrix} + \begin{pmatrix} P \\ Q \end{pmatrix}$$

$$\begin{pmatrix} \Pi\,S \\ 0 \end{pmatrix} + \begin{pmatrix} -A\Pi - B\,\Gamma \\ -C_s\,\Pi \end{pmatrix} \;=\; \begin{pmatrix} P \\ Q \end{pmatrix}$$

$$\begin{pmatrix} I_4 & 0 \\ 0 & 0_1 \end{pmatrix} \cdot \begin{pmatrix} \Pi \\ \Gamma \end{pmatrix} \cdot S + \begin{pmatrix} -A & -B \\ -C_s & 0 \end{pmatrix} \cdot \begin{pmatrix} \Pi \\ \Gamma \end{pmatrix} \cdot I_{3\times 3} \;=\; \begin{pmatrix} P \\ Q \end{pmatrix}$$

Once these matrices are known, we can determine the matrix

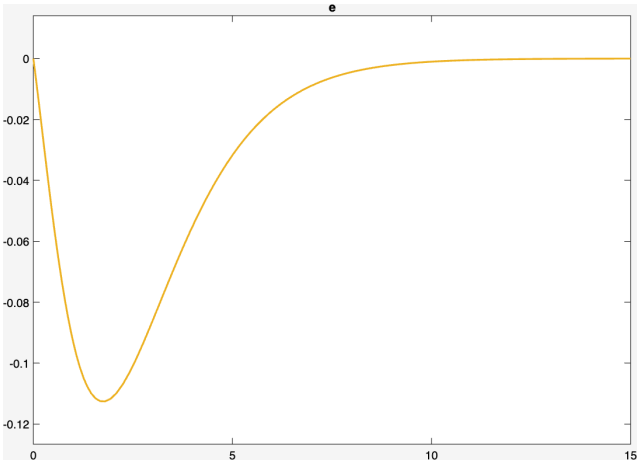
$$L \;=\; \Gamma - K\,\Pi$$

8 $\omega = 0.1$

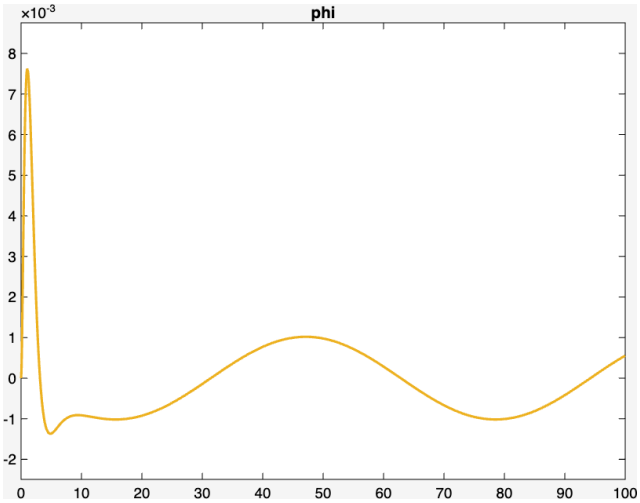
8.1 Linear system

For $\omega = 0.1$, the results are as follows.

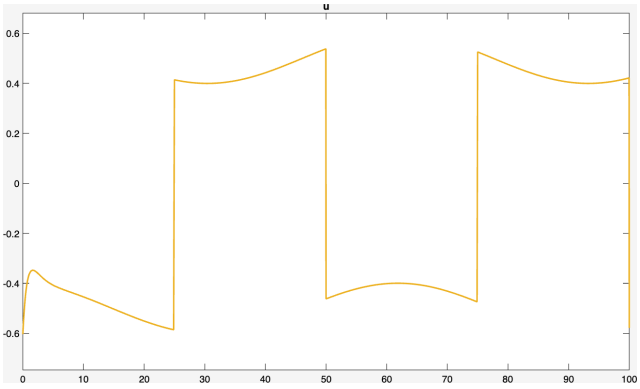
- The tracking error has an undershoot of -0.13 and converges in approximately $12s$.



- The pendulum angle oscillates around zero with an amplitude of 1 after an overshoot of 7.5.



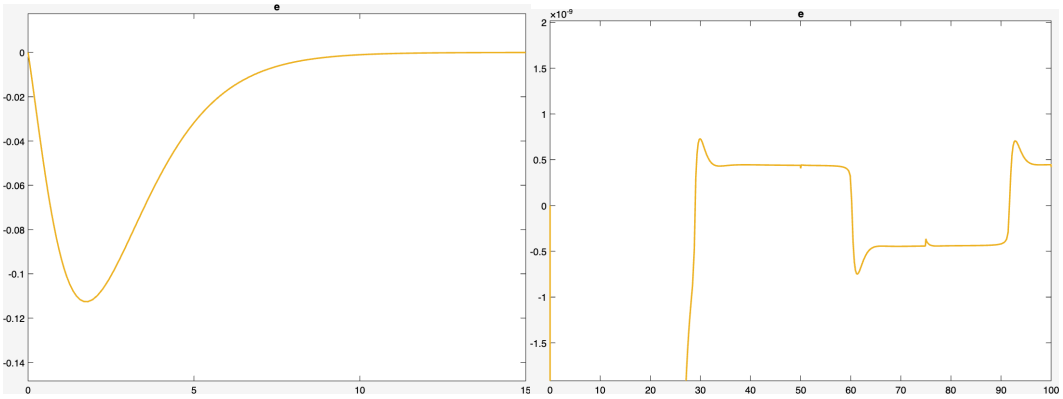
- The control input changes sign approximately every $25s$, in line with the transitions of the square wave d_1.



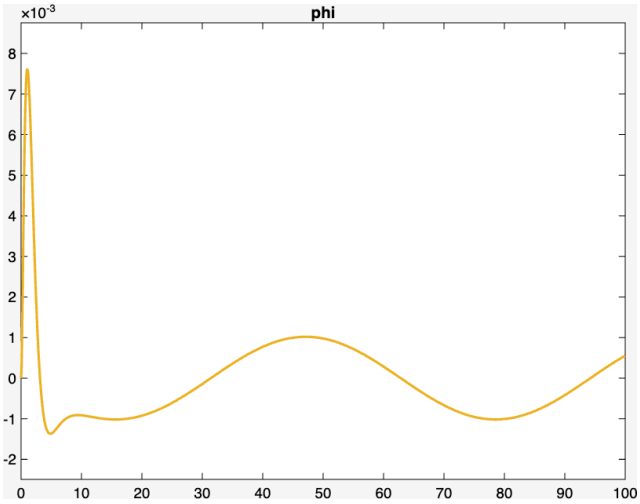
8.2 Nonlinear system

For $\omega = 0.1$, the results are as follows.

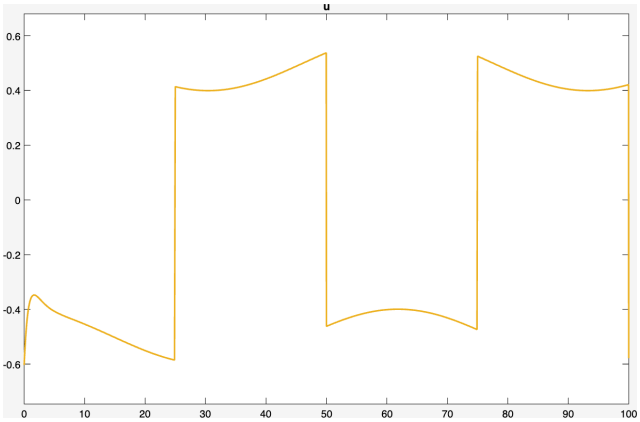
- The tracking error has an undershoot of -0.13 and continues to oscillate around zero in steady state.



- The pendulum angle oscillates around zero.



- The control input changes sign approximately every $25s$, in line with the transitions of the square wave d_1.

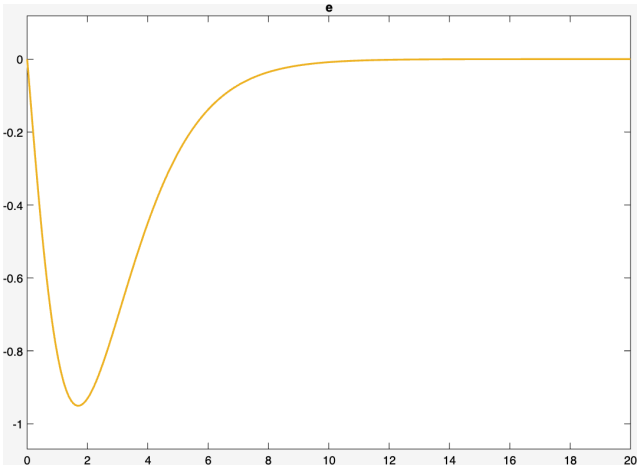


9 $\omega = 1$

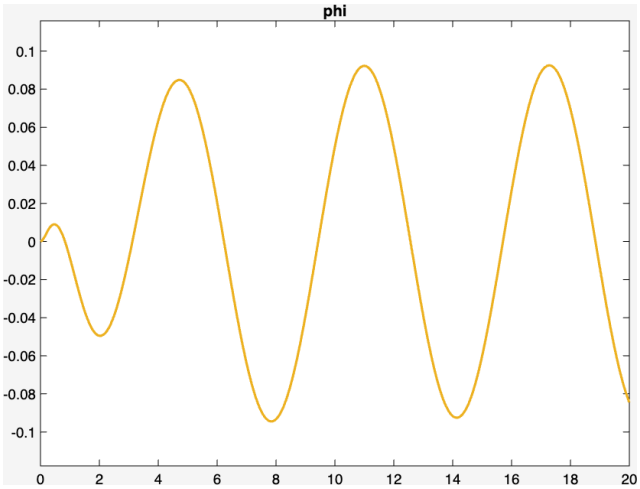
9.1 Linear system

For $\omega = 1$, the results are as follows.

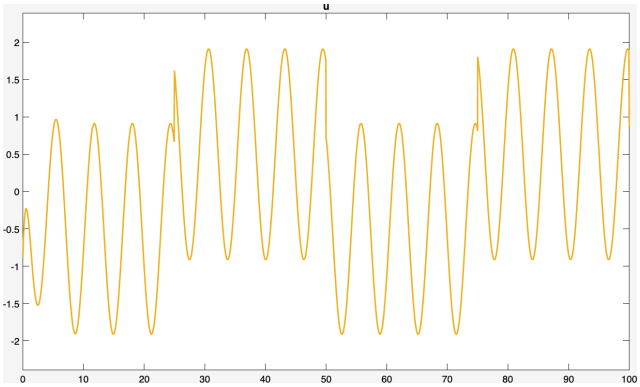
- The tracking error has an undershoot of -0.93 and converges in approximately $12s$.



- The pendulum angle oscillates around zero with a maximum amplitude of approximately 0.1.



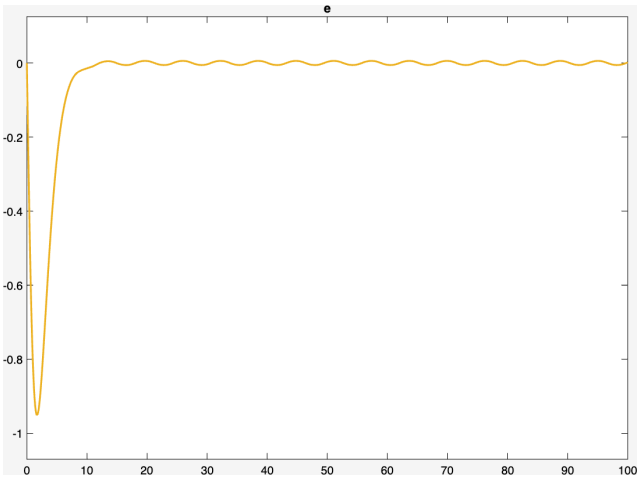
- The control input is smoother than in the previous case, but changes sign much more rapidly.



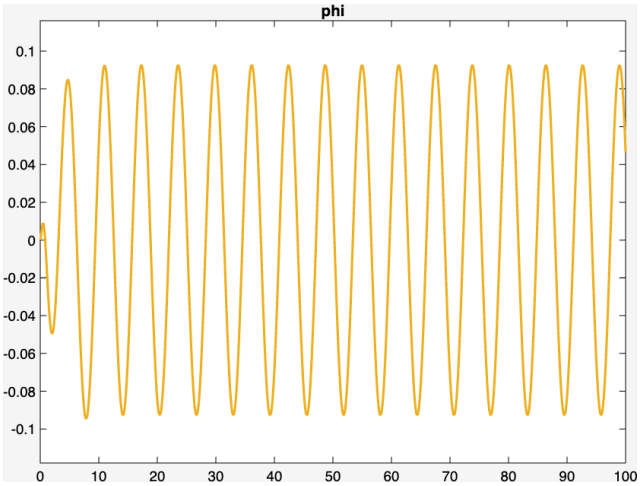
9.2 Nonlinear system

For $\omega = 0.1$, the results are as follows.

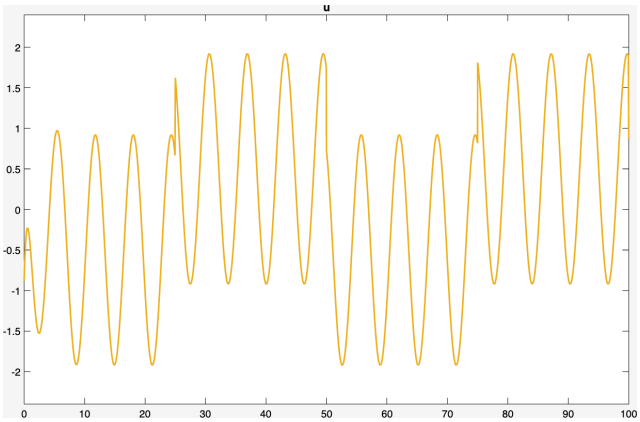
- The tracking error has an undershoot of -0.13 and continues to oscillate around zero in steady state.



- The pendulum angle oscillates around zero with an amplitude of 0.1.



- The disturbance d_1 affects the control input more strongly than in the linear case.

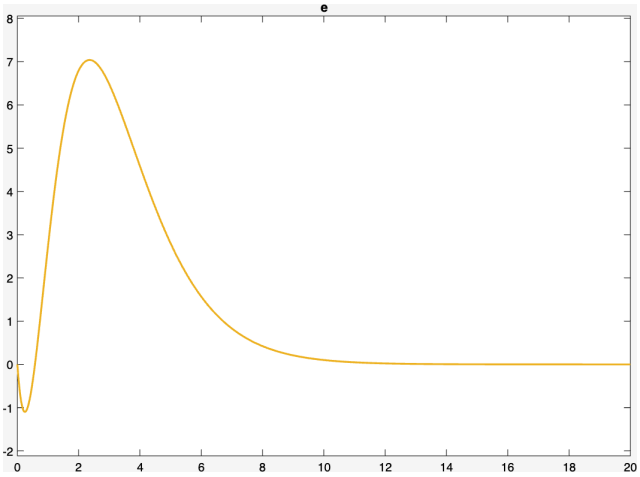


10 $\omega = 10$

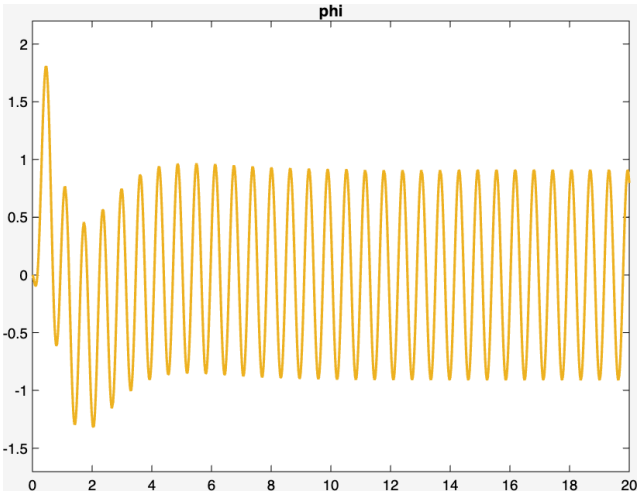
10.1 Linear system

For $\omega = 10$, the results are as follows.

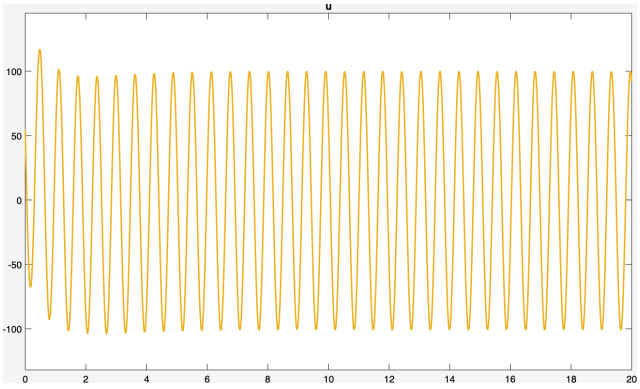
- The tracking error has an overshoot of 7 and converges in approximately $12s$.



- The pendulum angle oscillates around zero with a maximum amplitude of $1c.a.$



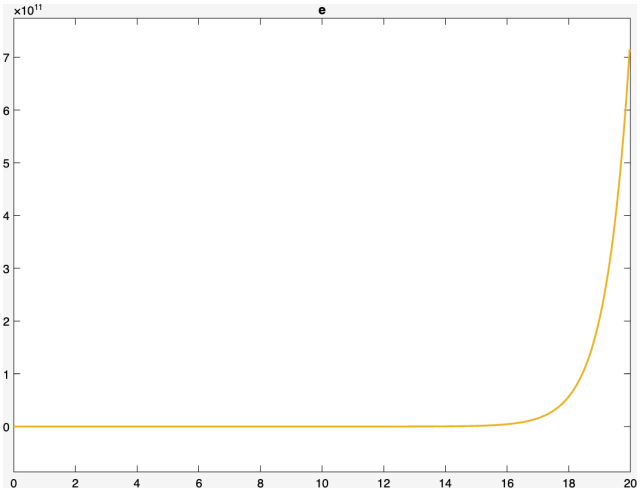
- The control input oscillates around zero with an amplitude of 100.



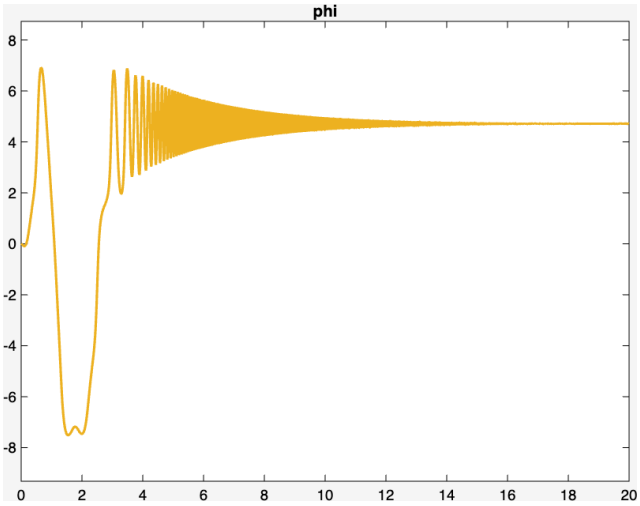
10.2 Nonlinear system

For $\omega = 0.1$, the results are as follows.

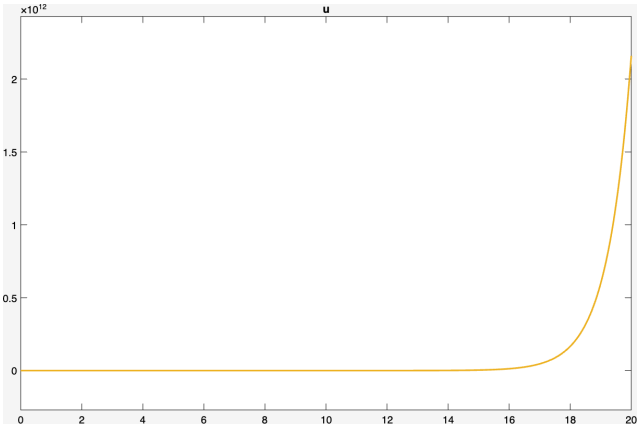
- The tracking error diverges.



- The pendulum angle oscillates around zero in steady state.



- The control input diverges.



11 Conclusions

The simulations yielded the following observations.

- **in the linear system, the disturbance $d_1(t)$ does not affect the output $y(t)$**. By the internal model principle, the controller rejects the disturbance, preventing it from affecting the tracking error.

- **in the nonlinear system, the regulation objective is achieved only approximately**. As long as the pendulum oscillations remain small, the linear approximation is accurate and the tracking error stays close to zero. As the angular frequency ω increases, both the oscillation amplitude and the required control effort increase, making the linear approximation less accurate. **At sufficiently high values of ω** (for example, $\omega = 10$), the difference between the linear model and the actual system can compromise stability and lead to **divergence of the tracking error and the control input**.